

# Euclid's Elements — Book I

## Proposition 24

CGJteamLab

*If two triangles have two sides equal to two sides respectively, but have one of the angles contained by the equal straight lines greater than the other, they will also have the base greater than the base.*

### 1 Introduction

Euclid's Proposition I.24 is the classical hinge theorem. Two triangles have two corresponding pairs of congruent sides. If the included angle of the first triangle is greater than the included angle of the second, then the side opposite the greater angle is greater than the corresponding side of the second triangle.

In the notation of the formal development, the hypotheses are

$$AB \cong DE, \quad AC \cong DF,$$

and

$$\angle EDF < \angle BAC.$$

The conclusion is

$$EF < BC.$$

The final Lean theorem therefore has the form

```
theorem euclid_proposition_24
  [HilbertIncidence Geo]
  [HilbertCongruence Geo]
  (A B C D E F : Geo.Point)
  (hABC : Not (PrimCollinear Geo A B C))
  (hDEF : Not (PrimCollinear Geo D E F))
  (hAB_DE : Geo.Congruent A B D E)
  (hAC_DF : Geo.Congruent A C D F)
  (hAngle :
    HilbertAngleLess Geo E D F B A C) :
  HilbertSegmentLess Geo E F B C := by
  ...
```

The reconstruction is entirely synthetic. No numerical length function, coordinate system, or numerical angle measure is introduced.

## 2 The Classical Difficulty

Euclid’s proof is considerably more delicate than the diagram suggests. The traditional construction transfers one of the given sides into the larger angle and then compares the resulting auxiliary triangle with the second given triangle. The difficulty is that the auxiliary point may occupy different positions relative to the opposite side of the first triangle.

A proof which silently assumes one position from the drawing is therefore incomplete. In a Hilbert-style formalization this issue cannot be hidden: the possible orders of the auxiliary points must be separated explicitly.

This becomes the central structural feature of the Lean proof of I.24.

## 3 Formal Strategy

The proof is divided into two principal stages.

First, an auxiliary point  $H$  is constructed on an interior ray of  $\angle BAC$  so that

$$AH \cong AC$$

and

$$BH \cong EF.$$

Thus the original comparison  $EF < BC$  is reduced to

$$BH < BC.$$

Second, the position of the ray through  $H$  relative to the side  $BC$  is analyzed. If the ray meets  $BC$  at  $Y$ , the points  $A, H, Y$  can occur in three possible orders:

$$A - H - Y, \quad H = Y, \quad A - Y - H.$$

The two nontrivial configurations are proved separately. The middle case is immediate from betweenness.

This decomposition is reflected in the main helper theorems:

```
euclid_proposition_24_sas_aux
euclid_proposition_24_hinge_case1
euclid_proposition_24_hinge_case3
euclid_proposition_24_hinge_aux
```

## 4 The SAS Construction

The theorem `euclid_proposition_24_sas_aux` performs the congruence part of the argument.

Starting from

$$AB \cong DE, \quad AC \cong DF, \quad \angle EDF < \angle BAC,$$

the strict angle comparison supplies an interior ray of  $\angle BAC$ . A point  $H$  is laid off on this ray with

$$AH \cong AC.$$

Since  $AC \cong DF$ , congruence transport also gives

$$AH \cong DF.$$

The angle at  $A$  cut out by the auxiliary ray is congruent to  $\angle EDF$ . Hence SAS applies to the auxiliary triangle  $ABH$  and the given triangle  $DEF$ . The corresponding third sides are congruent:

$$BH \cong EF.$$

The helper therefore packages the geometrically useful output:

- an interior ray of  $\angle BAC$ ;
- a point  $H$  on that ray;
- $AH \cong AC$ ;
- $BH \cong EF$ .

The remainder of I.24 is consequently a pure hinge comparison inside the triangle  $ABC$ .

## 5 The Hinge Configuration

Let the interior ray containing  $H$  meet the open segment  $BC$  at  $Y$ . The proof must establish

$$BH < BC.$$

The relation “same ray” does not by itself determine the order of  $H$  and  $Y$ . The formal proof therefore invokes the same-ray trichotomy and splits into three cases.

### 5.1 Case 1: $A - H - Y$

This is the longer branch. The point  $H$  lies between  $A$  and the intersection point  $Y$ .

The proof first reconstructs all required noncollinearity statements. It then uses Proposition I.16 twice. Together with the isosceles-triangle relation

$$AH \cong AC,$$

these exterior-angle comparisons produce a strict comparison between an angle at  $C$  and an angle at  $H$ .

The intersection point  $Y$  lies between  $B$  and  $C$ , so the ray  $HY$  is an interior ray of the angle at  $H$ . The theorem `hilbert_interior_angle_less` converts this incidence fact into a strict angle inequality. After transporting angles along the same rays, the proof obtains

$$\angle BCH < \angle BHC.$$

Proposition I.19 can now be applied to triangle  $BHC$ :

$$\angle BCH < \angle BHC \implies BH < BC.$$

Thus the first nontrivial order configuration is closed entirely by previously established synthetic angle theory.

## 5.2 Case 2: $H = Y$

If the auxiliary point itself lies on  $BC$ , then

$$B - H - C.$$

The definition of strict segment comparison immediately gives

$$BH < BC.$$

No angle argument is required.

## 5.3 Case 3: $A - Y - H$

In the third configuration the intersection point  $Y$  lies between  $A$  and  $H$ . This branch is shorter than Case 1.

At  $C$ , the ray  $CY$  meets the open segment  $HA$ . Therefore

$$\angle HCY < \angle HCA.$$

Since  $C - Y - B$ , the rays  $CY$  and  $CB$  coincide, so this comparison can be transported to the angle determined by  $CB$ .

Next,  $AH \cong AC$  makes triangle  $AHC$  isosceles. Hence its base angles are congruent. This transports the preceding strict inequality from the angle at  $C$  to the corresponding angle at  $H$ .

At  $H$ , the ray  $HY$  meets the open segment  $CB$ , giving another proper interior-angle comparison. Since  $H - Y - A$ , the rays  $HY$  and  $HA$  coincide. Combining the two strict inequalities again yields

$$\angle BCH < \angle BHC.$$

A final application of Proposition I.19 gives

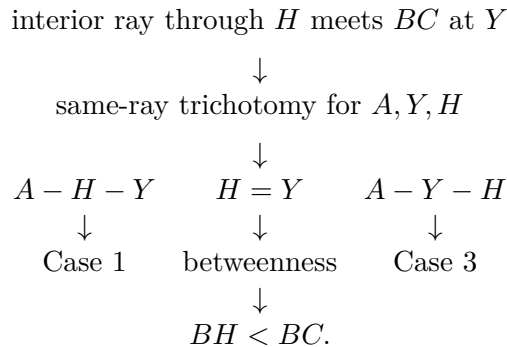
$$BH < BC.$$

Thus both nontrivial point-order configurations have the same final form, but they reach it through different angle-comparison paths.

## 6 The Hinge Helper

The theorem `euclid_proposition_24_hinge_aux` hides the three-way order analysis from the public theorem.

Schematically,



The important point is that no location of  $H$  is inferred from the diagram. The order structure supplies an exhaustive logical split.

## 7 Completion of Proposition I.24

After the two helpers have been established, the public theorem is short. The SAS construction supplies an auxiliary point  $H$  satisfying

$$BH \cong EF.$$

The hinge helper supplies

$$BH < BC.$$

Strict segment comparison is then transported through congruence, yielding

$$EF < BC.$$

The final architecture is therefore

$$\begin{array}{c}
 AB \cong DE, \quad AC \cong DF, \quad \angle EDF < \angle BAC \\
 \downarrow \\
 \text{euclid\_proposition\_24\_sas\_aux} \\
 \downarrow \\
 AH \cong AC, \quad BH \cong EF \\
 \downarrow \\
 \text{euclid\_proposition\_24\_hinge\_aux} \\
 \downarrow \\
 BH < BC \\
 \downarrow \\
 \text{segment-inequality transport through congruence} \\
 \downarrow \\
 EF < BC.
 \end{array}$$

## 8 Hilbert and Book Zero Components Used

### 8.1 Strict angle comparison

The relation `HilbertAngleLess` is essential. It represents a strict comparison synthetically: one angle is realized as a proper interior subangle of another. No real-valued angle measure is required.

### 8.2 Interior ray and crossbar machinery

The proof repeatedly turns ray–segment intersection into strict angle comparison. This is the formal replacement for statements which are usually read directly from a diagram.

### 8.3 Same-ray trichotomy

The trichotomy used in the hinge helper is not cosmetic. It is precisely what prevents the proof from assuming an unjustified location of the auxiliary point. The three cases record the complete order information hidden in the classical figure.

### 8.4 Isosceles base angles

The congruence

$$AH \cong AC$$

is used through `hilbert_isosceles_base_angles`. It provides the angle transport linking the two sides of the hinge configuration.

### 8.5 Propositions I.16 and I.19

Case 1 uses Proposition I.16 to create the required strict exterior-angle comparisons. Both nontrivial cases ultimately use Proposition I.19: the greater angle in a triangle is subtended by the greater side.

Thus I.24 is no longer an isolated argument. It is built from the angle order theory developed in the immediately preceding propositions.

## 9 Formalization Notes

Several features of the Lean proof deserve emphasis.

First, the dominant technical cost is not SAS. Once the auxiliary triangle has been constructed, SAS immediately gives  $BH \cong EF$ . The difficult part is the order geometry of the auxiliary point.

Second, noncollinearity must be transported through many permutations of `PrimCollinear`. These obligations are logically elementary but are indispensable because strict angle comparison is defined only for nondegenerate angles.

Third, equality of rays does not permit informal replacement of angle arguments. Congruence of the corresponding angles must be constructed and then strict inequalities transported explicitly.

Fourth, the proof makes visible a genuine structural distinction between Case 1 and Case 3. They are not merely syntactic variants. The first requires two exterior-angle comparisons from I.16, whereas the second can be organized directly through two interior-subangle comparisons and the isosceles base-angle theorem.

Finally, the completed proof introduces no new geometric axiom. It uses the Hilbert incidence, order, and congruence framework together with the Book Zero proof language and the already established propositions of Book I.

## 10 Comparison with Euclid’s Proof

The formal reconstruction preserves the geometric content of the hinge argument but changes its logical presentation.

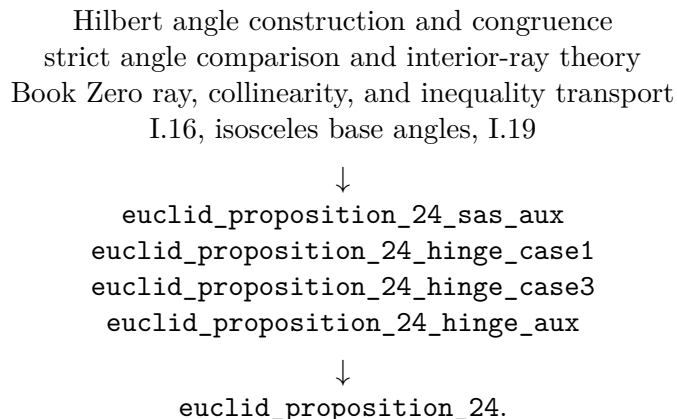
Euclid’s diagram encourages the reader to regard the auxiliary point as having one visually evident position. The formal proof instead treats its position as data to be proved. The same-ray trichotomy forces the missing case distinction into the proof itself.

This is an important example of what formalization contributes to the historical analysis of the *Elements*. The issue is not that the synthetic method is inadequate. Rather, the diagram suppresses order and separation information which a complete axiomatic proof must state.

The Hilbert reconstruction makes that information explicit while remaining fully synthetic.

## 11 Dependency Summary

The essential dependency graph is



No circle continuity, numerical metric, or additional hinge axiom is used.

## 12 Conclusion

Proposition I.24 is one of the most substantial order-sensitive arguments encountered so far in the reconstruction of Book I.

Its final statement is simple:

$$AB \cong DE, \quad AC \cong DF, \quad \angle EDF < \angle BAC \implies EF < BC.$$

But formalizing the classical hinge argument exposes the point-order information hidden by the traditional diagram.

The completed Lean proof resolves this by separating construction, congruence, order analysis, and angle comparison into reusable helpers. In particular, the explicit split into  $A - H - Y$ ,  $H = Y$ , and  $A - Y - H$  removes the ambiguity of the auxiliary configuration rather than assuming one case from the figure.

The complete development through Proposition I.24 passes `lake build`. Proposition I.24 introduces no new geometric axiom.